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LABOUR SUPPLY IN ITALY: AN EMPIRICAL ANALYSIS OF JOINT HOUSEHOLD DECISIONS, WITH TAXES AND QUANTITY CONSTRAINTS

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SUMMARY

This study applies an econometric framework which allows for complex non-convex budget sets, highly non-linear labour supply curves and imperfect markets with institutional constraints. A married couple's version of the model is estimated on Italian microdata. The empirical results show that male labour supply is rather inelastic while labour supply among females, especially participation, is considerably more elastic. The elasticities depend strongly on household income. The largest elasticities are found for females living in poor households. The results of the tax simulations suggest that there are only modest labour supply responses from replacing the 1987 system by proportional taxes. Copyright © 1999 John Wiley & Sons, Ltd.

1. INTRODUCTION

The purpose of this paper is to examine the effects of taxation on labour supply and on the distribution of income in Italy. Our study draws on the framework introduced by Dagsvik (1994) and may be viewed as an extension of the model in Dickens and Lundberg (1993). Our approach implies a modelling and estimation strategy which is rather different from the one proposed and applied to the USA by Hausman and co-authors (cf. Hausman, 1979, 1980, 1981 and 1985; Burtless and Hausman, 1978; Hausman and Ruud, 1984), and adopted in other studies; for example, Blomquist (1983) for Sweden, Arrufat and Zabalza (1986) for the UK and Colombino and Del Boca (1990) for Italy. A general presentation of the 'Hausman approach' can be found in Moffit (1986).

The Hausman approach is essentially a generalization of a Tobit model which takes account of the fact that taxes and benefits generate a kinked, piecewise linear and even non-convex budget constraint. In that approach, the individual is free to choose his or her optimal position along the budget constraint. However, the structure of labour costs makes it less attractive in general to firms to offer contracts that allow for flexible work schedules. As a consequence, the set of choices available to the individual may be severely reduced, or alternatively the costs of searching for non-standard contracts are high. For critical assessments of other aspects of the Hausman approach we refer to Kapteyn *et al.* (1990) and MaCurdy *et al.* (1990).

Recently, there have been a few empirical studies of labour supply models which account for restrictions on available jobs and hours. Blundell *et al.* (1987) allow for involuntary employment. Ilmakunnas and Pudney (1990) allow for three alternative hours of work (non-participation,

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part-time and full-time working). Dickens and Lundberg (1993) assume that each individual faces a random number of job offers where each job has a fixed quantity of hours. The number of jobs are drawn from a binomial distribution with a fixed maximum of 10 jobs. Dickens and Lundberg motivate the restrictions on hours with a reference to a mechanism that mimics a market process in which a limited number of jobs is offered by employers. Like Dickens and Lundberg (1993) we assume offered hours to be drawn from a market distribution. Opposed to Dickens and Lundberg, we do not introduce a fixed maximum of jobs and further, we allow each job to be characterized by more than one attribute. These attributes are a fixed quantity of hours, a wage rate and non-pecuniary attributes. Offered wages and other job attributes are drawn from a market distribution. The individual is assumed to choose the job from his or her opportunity set that maximizes his or her utility. It is important to note that the opportunity sets may vary across individuals. Some individuals have more opportunities than others due to differences in skills, education and age. To account for unobserved differences in preferences as well as in opportunity sets our framework allows for random taste shifters and random opportunity sets.

Van Soest (1995) offers an alternative approach where hours are discretized. The stochastic term which is added to the deterministic part of the utility function is assumed to be extreme value distributed. By assuming utility maximization van Soest thus obtains a multinomial logit model (see McFadden, 1978). However, van Soest notes that this model specification is unable to reproduce the observed data and adds dummies to the utility function in order to capture the peaks in the hours of work distribution.

Like van Soest (1995), we account for most details in the tax and benefit system when estimating the model. Dickens and Lundberg (1993) and Ilmakunnas and Pudney (1990), however, employ simplified versions of the true budget sets.

The present study is based on data from the 1987 Survey of Household Income and Wealth (SHIW) conducted by the Bank of Italy, which at the moment of estimation was the only survey collecting detailed information on incomes and hours of work of each household member. The microeconomic analysis of labour supply in Italy has been severely limited by the lack of nationwide data sets with relevant information to compute wage rates before 1987. The situation was rather paradoxical since the Survey of Labor Force conducted by the Italian Central Bureau of Statistics included hours of work but not incomes, while SHIW included incomes but not hours. Analysts had to resort to local samples or to the matching of different samples of households.

Structural analyses of labor supply in Italy based on microdata have been carried out by Colombino and Zabalza (1982), Colombino (1985), Colombino and Del Boca (1990), Del Boca and Flinn (1984) and Rettore (1990). The first three of these studies are based on a local sample of 1000 households. Colombino and Zabalza (1982) and Colombino (1985) estimate a CES utility function and model quantity constraints and disequilibrium by using specific information upon the willingness to work more or less, in a way similar to Ham (1982). In the first of these studies gross wages are used, while in the second taxes are accounted for by using a continuous differentiable approximation to the piecewise-linear budget. Colombino and Del Boca (1990) use the Hausman approach mentioned above and report estimates of the behavioural and welfare effects of the current as well as of alternative tax systems. All these analyses ignore those features of the tax system which might imply non-convexities of the budget set. Del Boca and Flinn (1984) and Rettore (1990) use particular interpretations of the participation condition to account for hours constraints, but ignore taxes. Most of the previous analyses focus on the wife's labour supply. None of them models both spouses' joint decisions.

To summarize, this paper presents the first model of labour supply in Italy which takes simultaneously into account.

- Both spouses' choices
- Exact representation of income taxes
- Constraints on the distribution of hours

and uses a nationwide sample of microdata.

The paper is organized as follows. Section 2 explains the model. Section 3 describes the data used to estimate the model. Section 4 deals with the employed empirical specification and discusses the estimation results. Section 5 presents the results of various tax simulations.

2. THE MODEL

For expository reasons we start with focusing on one-person households. Later in this section we outline how the model can be extended to the case of married couples.

Assume that the choice environment of an individual agent consists of a set of market (jobs) and non-market opportunities. With each market opportunity is associated a fixed number of hours of work, a wage rate and other job characteristics. For non-market opportunities hours of work as well as the wage rate are equal to zero. Let $B_i(h, w)$ denote the set of market opportunities with hours of work $h > 0$ and wage rate $w > 0$ and $B_i(0, 0)$ the set of non-market opportunities for household i .

For each household the budget constraint is given by

$$C = f(wh, I) \quad (1)$$

where C is after-tax income, I is non-labour income and $f(\cdot)$ is a function that transforms gross income into disposable income. This function is uniquely determined by the tax and benefit rules.

The household has preferences over hours of work, income and non-pecuniary attributes. Let $U_i(C, h, j)$ be the utility of household i when opportunity $j \in B_i(h, w)$ is chosen. Note that when j is chosen, hours of work and income follow. We assume that the utility has the form

$$U_i(C, h, j) = v_i(C, h)\varepsilon_{ij} \quad (2)$$

where $v_i(\cdot, \cdot)$ is a positive deterministic term, and ε_{ij} is a random taste shifter that is supposed to capture the effect of unobservable attributes associated with opportunity j . Note that the term ε_{ij} is viewed as random solely from the econometrician's point of view, while it is assumed known to the household. Specifically, this term accounts for the fact that for a given agent, tastes may vary across opportunities (wages, hours and other non-pecuniary attributes), and for a given opportunity, tastes may vary across agents. Assume that the taste shifters are i.i.d. with c.d.f.

$$P(\varepsilon_{ij} \leq y) = \exp\left(-\frac{1}{y}\right), \quad y > 0 \quad (3)$$

Now, let

$$\Psi_i(h, w) = v_i(f(wh, I), h) \quad (4)$$

where we have inserted for equation (1) in the structural term $v_i(\cdot, \cdot)$, and let

$$\eta_i(h, w) = \max_{j \in B_i(h, w)} \varepsilon_{ij} \quad (5)$$

The household is assumed to maximize the utility in equation (2) under the budget constraints (1) and given the opportunity set available for the household. Then we can write

$$\max_{j \in B_i(h, w)} U_i(f(wh; I), h, j) = \Psi_i(h, w) \eta_i(h, w) \quad (6)$$

To simplify the notation let $z = (h, w)$ and suppress the dependency on I . Moreover, let $M_i(z)$ be the number of opportunities in $B_i(z)$. For expository simplicity we shall assume that the total set of possible hours and wages is finite. From equations (3) and (5) we then get the following expression for the probability distribution of $\eta_i(z)$:

$$\begin{aligned} \Pr(\eta_i(z) \leq y) &= \Pr\left(\max_{j \in B_i(z)} \varepsilon_{ij} \leq y\right) \\ &= \prod_{j \in B_i(z)} \Pr(\varepsilon_{ij} \leq y) = \prod_{j \in B_i(z)} \exp\left(-\frac{1}{y}\right) = \exp\left(-\frac{M_i(z)}{y}\right) \end{aligned} \quad (7)$$

Accordingly, the distribution F of $\Psi_i(z)\eta_i(z)$ is given by

$$F(y) = \Pr(\Psi_i(z)\eta_i(z) \leq y) = \exp\left(-\frac{\Psi_i(z)M_i(z)}{y}\right) \quad (8)$$

In order to derive the probability $\varphi_i(z)$ that individual i chooses a job with $z = \tilde{z}$, we use the fact that $\varphi_i(\tilde{z})$ can be expressed in the following way:

$$\varphi_i(\tilde{z}) = \Pr\left(\Psi_i(\tilde{z})\eta_i(\tilde{z}) \geq \max_{z \neq \tilde{z}} (\Psi_i(z)\eta_i(z))\right) \quad (9)$$

Now, let

$$V_i = \max_{z \neq \tilde{z}} (\Psi_i(z)\eta_i(z)).$$

Then we get from equation (8) that the distribution G of V_i is given by

$$G(y) = \Pr(V_i \leq y) = \prod_{z \neq \tilde{z}} \Pr(\Psi_i(z)\eta_i(z) \leq y) = \exp\left(-\frac{\sum_{z \neq \tilde{z}} \Psi_i(z)M_i(z)}{y}\right) \quad (10)$$

Since $\psi(\tilde{z})\eta(\tilde{z})$ and V_i are independent random variables with c.d.f. F and G , respectively, we get

$$\varphi_i(\tilde{z}) = \Pr(\Psi_i(\tilde{z})\eta_i(\tilde{z}) \geq V_i) = \int_0^\infty G(y) dF(y) \quad (11)$$

By inserting equations (8) and (10) in (11) we obtain the following expression for $\varphi_i(\tilde{z})$:

$$\varphi_i(\tilde{z}) = \Psi_i(\tilde{z})M_i(\tilde{z}) \int_0^\infty \exp\left(-y \sum_z \Psi_i(z)M_i(z)\right) dy = \frac{\Psi_i(\tilde{z})M_i(\tilde{z})}{\sum_z \Psi_i(z)M_i(z)} \quad (12)$$

An alternative expression can be derived from introducing the following fractions:

$$g_i(h, w) = \frac{M_i(h, w)}{\sum_{x>0} \sum_{y>0} M_i(x, y)} \quad (13)$$

and

$$g_{0i} = \frac{\sum_{x>0} \sum_{y>0} M_i(x, y)}{M_i(0, 0)} \quad (14)$$

Note that $g_i(h, w)$ is the number of jobs with characteristics $\{h, w\}$ relative to the number of feasible jobs, while g_{0i} is the number of feasible jobs relative to the number of non-market opportunities. Thus, $g_i(\cdot)$ and g_{0i} give a formal representation of the B_i sets introduced above.

From equations (12)–(14) we get

$$\varphi_i(h, w) = \frac{\Psi_i(h, w)g_{0i}g_i(h, w)}{\Psi_i(0, 0) + g_{0i} \sum_{x>0} \sum_{y>0} \Psi_i(x, y)g_i(x, y)} \quad (15)$$

for $\{h, w\} > 0$, and

$$\varphi_i(0, 0) = \frac{\Psi_i(0, 0)}{\Psi_i(0, 0) + g_{0i} \sum_{x>0} \sum_{y>0} \Psi_i(x, y)g_i(x, y)} \quad (16)$$

for $h = w = 0$.

Expressions (15) and (16) for the labour supply density $\varphi_i(h, w)$ turn out to be particularly attractive since they are functions of the structural term of the utility function, $\Psi_i(\cdot)$, the number of feasible jobs relative to non-market opportunities, g_{0i} , and the distribution of feasible market opportunities, $g_i(\cdot)$.

Both the preference term and the market opportunities may differ across individuals as is also indicated by the subscript i in equations (15) and (16). If individual differences in the opportunity sets are ignored, $g(h, w)$ may be interpreted as the fraction of jobs with hours h and wage rate w available in the economy.

An important issue is the identification of the preference term and the opportunity densities appearing in the labour supply densities which are derived from the model defined by equations (1)–(3). In an economy where there are as many market opportunities as there are non-market opportunities, $g_0 = 1$. We will call this state of the world ‘a fair environment’. When the market opportunities are less than the non-market opportunities, $g_0 < 1$. We may interpret

this situation as a state where the number of jobs is less than in the benchmark state of 'a fair environment'.

When individuals face a fair environment, we get the probability, $\tilde{\varphi}(0, 0)$, for non-participation:

$$\tilde{\varphi}(0, 0) = \frac{\Psi(0, 0)}{\Psi(0, 0) + \sum_{x > 0} \sum_{y > 0} \Psi(x, y)g(x, y)} \quad (17)$$

Unemployment can be associated with a situation where there are too few market opportunities in the opportunity sets. Accordingly, the unemployment rate, ρ , may be defined as the rate of relative excess non-participation with respect to the benchmark of a fair environment,

$$\rho = \frac{[1 - \tilde{\varphi}(0, 0)] - [1 - \varphi(0, 0)]}{1 - \tilde{\varphi}(0, 0)} \quad (18)$$

Inserting equations (16) and (17) for $\varphi(0, 0)$ and $\tilde{\varphi}(0, 0)$ into (18) we get

$$\rho = (1 - g_0)\varphi(0, 0) \quad (19)$$

Thus, g_0 is non-parametrically identified from observations of the unemployment rate (ρ) and the participation rate ($1 - \varphi(0, 0)$).

Assume that the deterministic part of the utility function is separable in income and leisure, that is,

$$v(C, h) = v_1(C)v_2(h) \quad (20)$$

From equations (1), (4), (15), (16) and (20) we get

$$\ln \frac{\varphi(h, w, I, k)}{\varphi(0, 0, I, k)} = \ln v_1(\tilde{f}(wh, I, k)) - \ln v_1(f(0, I)) + \ln v_2(h) - \ln v_2(0) + \ln g_0 g(w, h) \quad (21)$$

where k is fixed costs of working and $\tilde{f}$ is the tax function given fixed costs of working. Above, fixed costs of working have been subsumed under the functional form f .

Now we can use the variation of the f -function with k to identify $v_1(\cdot)$ non-parametrically. From equation (21) we get

$$\frac{\partial \ln \frac{\varphi(h, w, I, k)}{\varphi(0, 0, I, k)}}{\partial k} = \frac{\frac{\partial v_1(f(hw, I, k))}{\partial f}}{v_1(f(wh, I, k))} \cdot \frac{\partial f(wh, I, k)}{\partial k} \quad (22)$$

The left-hand side of equation (22) is observed and $\partial f/\partial k$ is observed. Thus equation (22) yields a differential equation for v_1 and consequently v_1 is identified.

It remains to identify $v_2(h)g(h, w)$. By noting that $g(h, w) = g_1(w|h)g_2(h)$, where $g_1(\cdot | \cdot)$ is the conditional density of wages and $g_2(\cdot)$ is the marginal density of hours, we get after some

straightforward elaboration on the ratio $\varphi(h, w)/\varphi(0, 0)$ (for the moment the dependency on I is suppressed):

$$g_1(w|h) = \frac{\varphi(h, w)/v_1(f(hw))}{\int \varphi(h, y)/v_1(f(hy)) \, dy} \quad (23)$$

Since $\varphi(\cdot)$ is observed and $v_1(\cdot)$ is identified from equation (22), $g_1(w|h)$ is identified.

To identify each term in the product $v_2(h)g_2(h)$ non-parametrically will require additional information on institutional variables or firm-specific variables that affect $g_2(h)$. Such information has, however, not been available. The identification of $v_2(h)$ and $g_2(h)$ thus requires parametric assumptions. The term $v_2(h)$ is identified up to a multiplicative constant when offered hours are assumed to be uniformly distributed, which means that there are no restrictions on offered hours. However, due to the technology within firms and negotiations between firms and unions we find it more realistic to assume that there are restrictions on offered hours in the market. Specifically, we will assume that hours are uniformly distributed except for a peak at full-time hours. For further details, see Section 4.

The extension of equations (15) and (16) to cover the case of married couples is straightforward and completely analogous to the case of single-person households. In this case the household has preferences over income, leisure for the husband and wife, and for other implicit or explicit characteristics of the opportunities.

Now let the subscripts M and F denote male and female. Let g_{00} , g_{0F} and g_{0M} be the analogue to g_0 in the one-person household case and let $g(h_F, h_M, w_F, w_M)$ be the analogue to $g(h, w)$. The analogue to equation (15) is (subscript i indicating household i is suppressed):

$$\varphi(h_F, h_M, w_F, w_M) = \frac{\Psi(h_F, h_M, w_F, w_M)g(h_F, h_M, w_F, w_M)}{D} \quad (24)$$

for $h_M > 0$, $h_F > 0$, $w_M > 0$, $w_F > 0$ and where

$$\begin{aligned} D = & g_{0F}g_{0M} \sum_{x_1 > 0} \sum_{x_2 > 0} \sum_{y_1 > 0} \sum_{y_2 > 0} \Psi(x_1, x_2, y_1, y_2)g(x_1, x_2, y_1, y_2) \\ & + g_{0F} \sum_{x_2 > 0} \sum_{y_2 > 0} \Psi(0, x_2, 0, y_2)g_M(x_2, y_2) \\ & + g_{0M} \sum_{x_1 > 0} \sum_{y_1 > 0} \Psi(x_1, 0, y_1, 0)g_F(x_1, y_1) + g_{00}\Psi(0, 0, 0, 0) \end{aligned} \quad (25)$$

where $g_M(\cdot)$ and $g_F(\cdot)$ are marginals of g . The expression for $\varphi(0, h_M, 0, w_M)$, $\varphi(h_F, 0, w_F, 0)$ and $\varphi(0, 0, 0, 0)$ are straightforward analogues of equation (24).

For the sake of simplicity and intuition we have presented a discrete version of the model. It should be noted that the model is conceived as a continuous one, i.e. h and w can vary continuously. The sums in equation (25) are replaced by integrals, and the functions g are replaced by densities. For further details we refer to Dagsvik (1994) and Aaberge *et al.* (1995).

3. DATA

The estimation of the model is based on data from the 1987 SHIW. This survey is now conducted every two years by the Central Bank of Italy and contains detailed information on incomes and wealth of each household component. Moreover, it contains information on some socio-demographic characteristics. (The basic sampling design is a two-stage sample with stratification in the first stage.) We use the 1987 survey because at the moment of estimation it was the first and only one collecting hours of work, thus allowing to compute individual wage rates. The labour incomes measured by the survey are *net* of social security contributions and of taxes on personal income. Therefore, in order to obtain gross incomes we have to apply the 'inverse' tax code. Hourly wage rates are derived on the basis of gross annual wage income and observed hours. Detailed tax and benefit rules are accounted for in the derivation of household disposable income. The most distinctive feature of the tax and benefit system is the individual-based income tax. In April 1977 joint taxation of husband(s) and wife(s) was abolished and since then spouses are entitled to separate assessment. Individual income comprises incomes from any sources, excluding a few—such as interest—which are taxed according to a different code. Various allowances can be subtracted from individual income before applying tax rates. Among allowances we can only take into account compulsory social security contributions. Other major allowances (usually up to a ceiling) are: certain medical expenditures, passive interests, secondary and university education expenditures, some voluntary insurances. The survey does not contain sufficient information to include them in the model.¹

The following marginal tax rates, as of 1987, are applied to the tax base:

<i>Income (1000 LIT)</i>	<i>Marginal tax rate (%)</i>
up to 6000	12
6000–11,000	22
11,000–28,000	27
28,000–50,000	34
50,000–100,000	41
100,000–150,000	48
150,000–300,000	53
300,000–600,000	58
over 600,000	62

The tax rates above are national ones. In 1987 there were no local tax rates of relevance for labour supply choices.

From the gross tax, tax credits can be subtracted (up to the amount of the gross tax). There are two main types (in 1000 LIT):

- Credit for income production expenditures. Standard deduction for dependent workers is 492 plus a further 156 if taxable income does not exceed 11,000 (deductible fixed costs of working denoted k in equation (21)).

¹ It should be noted that most of those allowances depend upon household choices. Therefore it is not obvious that it would be better to take them into account, unless, of course, one is able to model those choices jointly with labour supply decisions.

- Credits for dependent family members. Assuming for simplicity that the husband is the primary earner, he is given a credit of 360 provided the wife's income does not exceed 3000. For every child or dependent family member the credit amount to 96. The credits for children are halved if the wife's income exceeds 3000.

There is a system of benefits transferred to families. The benefit amount depends on the number of household members and on the level of total gross family income. In 1987, a typical three-member household, was eligible to the following monthly benefits:

<i>Total gross yearly household income (1000 LIT)</i>	<i>Monthly benefit (1000 LIT)</i>
up to 12,000	160
12,001–15,000	140
15,001–18,000	110
18,001–21,000	80
21,001–24,000	50
24,001–27,000	20
over 27,000	0

Married couples with income from self-employment, which exceeds 20% of gross household income, have been excluded from the sample. The exclusion of families whose income is in large part due to self-employment is justified since their decision process may be substantially different from employees' and typically involves a permanent element of uncertainty in the decision process. Moreover, we do not observe hours for the self-employed. For the included households,

Table I. Sample values — married couples

	Mean	St. dev.	Min.	Max.
Individual variables:				
Annual hours of work				
Males	1995	544	0	3432
Females	690	909	0	3432
Participation rates (%)				
Males	96	18	—	—
Females	40	49	—	—
Hourly wage rates (1000 LIT)				
Males	12.0	6	0	63
Females	4.3	6	0	51
Gross annual earnings (1000 LIT)				
Males	24,700	13,900	0	13,900
Females	7,000	9,500	0	43,900
Age of wife	40.1	9.6	20	67
Age of husband	43.7	9.8	21	68
Household variables:				
Annual net taxes paid (1000 LIT)	9,600	6,600	0	59,000
Gross annual income (1000 LIT)	37,500	19,400	5600	142,500
Disposable annual income (1000 LIT)	27,900	13,300	5400	93,600
Number of children below 6	0.33	0.59	0	4
Number of children 6–13	0.58	0.78	0	4

observed income from self-employment has been added to net household income and it is treated as exogenous.

We have restricted the ages of the husband and wife to be between 20 and 68 years old. This age interval was chosen to make the estimates comparable to those obtained for other countries with the same type of model. We found, however, that the estimates show very little sensitivity to changes in the age interval. The sample covers 2953 households and summary statistics are presented in Table I.

4. EMPIRICAL SPECIFICATION AND ESTIMATION RESULTS

In this study we impose the following functional form on the deterministic part (4) of the utility function, where the subscripts F and M denote females and males:

$$\begin{aligned} \log \Psi(C, h_F, h_M) = & [\alpha_2(1 - K_M)(1 - K_F) + \alpha_3 K_F + \alpha_4 K_M] e^{\alpha_1 C} \\ & + [\alpha_6 + \alpha_7 \log A_M + \alpha_8 (\log A_M)^2] \left(\frac{L_M^{\alpha_5} - 1}{\alpha_5} \right) \\ & + [\alpha_{10} + \alpha_{11} \log A_F + \alpha_{12} (\log A_F)^2 + \alpha_{13} CU6 + \alpha_{14} CO6] \left(\frac{L_F^{\alpha_9} - 1}{\alpha_9} \right) \end{aligned} \quad (26)$$

where A_j is the age of gender j , $CU6$ and $CO6$ are number of children below and above 6 years old, C is household consumption and L_j is leisure for gender j , defined by

$$L_j = 1 - \frac{h_j}{8760}$$

$K_j = 1$ if spouse j is working; otherwise $K_j = 0$, and the specification implies that the marginal utility of consumption differs with respect to the reported labour market participation. The reason for doing this is the possible existence of non-reported income. The underground economy in Italy is believed to be of some importance. To capture some of these effects on income, the marginal utility of consumption is specified as shown above. Note that the presence of the K 's in the utility function leads to a non-separable structure in consumption and leisure.

The assumption of independence of offered hours and wages is standard in labour supply studies (an exception is Moffit, 1984). There are certainly many reasons to find plausible some dependence due to fixed labour costs, productivity varying with hours or selection of hours. While we do think that testing the independence assumption is an important and interesting line of research, this development is left for future work, and meanwhile the assumption is to be taken as a simplification. On the other hand, in Italy, as in other European countries, working hours are regulated by law or set in central negotiations between unions and employers' associations without any reference to the wage structure within firms. This fact weakens the assumed dependency between offered hours and offered wage rates and may justify the following independence assumption:

$$g(h_F, h_M, w_F, w_M) = g_1(w_F, w_M) g_2(h_F, h_M) \quad (27)$$

Moreover, we assume that feasible hours in the market is uniformly distributed except for peaks at full-time hours, which are defined by the interval [1846, 2106]. Note that this interval corresponds to weekly hours between 36 and 40. Thus,

$$g_2(h_F, h_M) = g_{2F}(h_F)g_{2M}(h_M) \quad (28)$$

with

$$g_{2k}(h) = \begin{cases} \gamma_k & \text{if } h \in [0, 1846) \\ \gamma_k \theta_k & \text{if } h \in [1846, 2106] \\ \gamma_k & \text{if } h \in (2106, 3432] \end{cases} \quad k = F, M \quad (29)$$

where 3432 is the maximum number of hours observed in the sample.

Moreover, we specify

$$\theta_F = \exp \alpha_{21} \quad (30)$$

and

$$\theta_M = \exp \alpha_{22} \quad (31)$$

The terms γ_F and γ_M cancel out in the likelihood, so that we estimate α_{21} and α_{22} (and therefore θ_F and θ_M). However, since g_{2F} and g_{2M} are defined to be probability densities we must also have that

$$\gamma_k = \frac{1}{3172 + 260\theta_k} \quad k = F, M \quad (32)$$

The fraction of market opportunities g_{0F} and g_{0M} are assumed to depend on local unemployment rates and on whether individuals are living in northern or southern Italy. The former is assumed to reflect how demand shocks may have an impact on employment opportunities. It is important to emphasize that the local unemployment rates are instruments or proxies for demand shocks variables. Their presence in g_{0k} does not follow from the model. Hence, g_{0F} and g_{0M} are given by

$$\log g_{0F} = \alpha_{15} + \alpha_{16} \text{Reg}_F + \alpha_{17} UE_F \quad (33)$$

and

$$\log g_{0M} = \alpha_{18} + \alpha_{19} \text{Reg}_M + \alpha_{20} UE_M \quad (34)$$

where $\text{Reg} = 1$ means living in northern regions (north of Tuscany) and 0 otherwise, and UE_k is the gender-specific local ratio of unemployed and employed individuals.

Note that positive α_{16} and α_{19} mean that living in northern Italy improves the chances of finding a market opportunity, compared to living in central and southern Italy. Likewise, negative values of α_{17} and α_{20} indicate that unemployment has a negative impact on job opportunities.

The density of offered wages is assumed to be lognormal with gender specific means that depend on length of schooling and on past potential working experience, where experience

is defined equal to age minus length of schooling minus six. Thus, the wage equations are given by

$$\log W_k = \beta_{0k} + \beta_{1k}s_k + \beta_{2k}Ex_k + \beta_{3k}(Ex_k)^2 + \beta_{4k}Reg_k + \xi_k \quad k = F, M \quad (35)$$

where (ξ_F, ξ_M) are normally distributed, s_k and Ex_k denote years of schooling and working experience.

The estimation is based on a procedure suggested by McFadden (1978) which yields results that are close to the full information maximum likelihood method. We are not able to use the exact likelihood function to estimate the model because the evaluation of the sums (integrals) in equation (25) would be too costly and cumbersome. The estimation procedure applied replaces the quadruple integral in the denominators of the densities by a sum over 70 random points, where each term is adjusted by appropriate weights. In other words, the continuous logit model is replaced by a discrete logit version. McFadden has demonstrated that this method yields consistent and asymptotically normal parameter estimates. We found the McFadden estimation procedure to be remarkably efficient. Our experience suggests that even choice sets of 10 random points (draws in R^4) produce results which are close to the one obtained by the 70 random point sets. The result of the estimations are reported in Tables II and III.

The estimates imply that the deterministic part of the utility function is an increasing and strictly concave function of leisure and consumption. The crucial parameters of the utility function are α_1 , α_5 and α_9 . These parameters are measured with good precision. The marginal

Table II. Estimates of the parameters of the utility function and of the opportunity density

Variables	Coefficients	Estimates	t-values
Consumption	α_1	$-0.780 \cdot 10^{-4}$	7.7
	α_2	-15.93	8.3
	α_3	-10.02	19.1
	α_4	-15.36	11.4
Male leisure	α_5	-18.65	16.4
	α_6	-0.1	1.4
	α_7	0.1	1.5
	α_8	-0.0	1.4
Female leisure	α_9	-6.8	8.1
	α_{10}	34.42	2.2
	α_{11}	-19.03	2.2
	α_{12}	2.7	2.3
	α_{13}	0.2	1.8
	α_{14}	0.2	2.7
Female opportunity density	α_{15}	-0.9	5.7
	α_{16}	0.7	6.5
	α_{17}	-0.5	0.9
Male opportunity density	α_{18}	-0.5	1.5
	α_{19}	0.3	1.2
	α_{20}	-0.2	0.1
Full-time peak, females	α_{21}	2.4	28.0
Full-time peak, males	α_{22}	2.5	51.9

Table III. Wage opportunity density: simultaneous ML estimates procedure versus OLS^a

Variables	Males		Females	
	OLS	Simultaneous ML	OLS	Simultaneous ML
Intercept	1.312 (33.2)	0.970 (17.6)	1.010 (16.9)	0.765 (10.2)
Education	0.058 (38.0)	0.063 (28.6)	0.077 (29.1)	0.086 (25.0)
Experience	0.030 (12.7)	0.035 (12.3)	0.025 (6.9)	0.024 (5.8)
Experience squared	$-0.397 \cdot 10^{-3}$ (-9.6)	$-0.440 \cdot 10^{-3}$ (-9.1)	$-0.307 \cdot 10^{-3}$ (-4.2)	$-0.234 \cdot 10^{-3}$ (-2.9)
Region	0.152 (13.2)	0.175 (12.7)	0.081 (4.2)	0.082 (3.5)
Standard error		0.301 (58.2)		0.327 (41.7)
R^2	0.39		0.44	

^a *t*-values in parentheses.

utility of leisure depends also on personal characteristics such as age and number of children. The estimates for the coefficients of these variables are less precise. Children have the expected positive effect on the value of wife's leisure. However, a rather surprising result is that the presence of older children have essentially the same effect as younger ones; as a matter of fact the point estimate for the former ones is even larger (this result, however, accords with other analyses of Italian data (see e.g. Colombino and Del Boca, 1990)). A possible explanation might be found in a cohort effect. Women with older children on average belong to older cohorts. For a variety of unobserved factors (attitudes, supply of child-care services, etc.) which change from one cohort to the other, older cohorts presumably tend to use a more 'leisure-intensive' technology in child-care.

The estimated parameters of the jobopportunities density confirm — at least for females — a more favourable environment in northern regions. On the other hand, the effect of unemployment is not measured precisely enough to draw any clear conclusion. Note that the estimates of α_{15} to α_{20} imply that g_{0j} , ($j = F, M$), is less than 1. Using expressions (29)–(32), we can derive the implications of the estimates of α_{21} and α_{22} . For example, the proportion of full-time jobs in the set of available market opportunities is $260 \exp(\alpha_{2j}) / (3172 + 260 \exp(\alpha_{2j}))$, $j = 1, 2$, which gives a proportion of 0.47 for women and 0.50 for men.

A remarkable result shown in Table III is that the wage densities estimated jointly with labour supply densities differ very little from OLS estimates. This result indicates that the selection effect is very weak. It should be noted that all coefficients are precisely determined and with signs as expected. Wages are strictly concave functions of experience with a maximum at 40 and 51 years of age for males and females, respectively. Wages in northern regions of Italy are estimated to be higher than in central and southern Italy. The regional effect is stronger for men than for women.

5. LABOUR SUPPLY ELASTICITIES

As part of the examination of the empirical model this section provides various labour supply elasticities.

The vast majority of labour supply studies report mean-sample elasticities. These elasticities do not capture the heterogeneity of the population and can lead to a misleading interpretation of the aggregate responses in the economy. As an alternative this study provides exact aggregate labour supply elasticities which account for observed and unobserved heterogeneity in preferences across the population.

Table IV reports aggregate uncompensated and compensated elasticities. The uncompensated elasticities are derived by predicting (by stochastic simulation) labour supply for each household (wife and husband) under the current regime and when the wage rates are increased by 1%. The aggregate uncompensated elasticity of, for example, female labour supply is obtained by calculating the relative change in the mean female labour supply (over all females in the sample) that results from a 1% wage increase. The 'estimates' in Table IV are based on 10 sets of simulations. Therefore, we have a distribution of 10 'estimates'. What we report in Table IV is the mean and standard deviation for each elasticity. The aggregate compensated elasticities are 'estimated' by the same procedure, but under the restriction that utility levels should remain unchanged. In all these simulations the opportunity and hours densities along with preferences are kept fixed. Note that the elasticities reported in Tables IV and V are computed on the basis of the demographic and educational structure, the tax system and the institutional constraints that the agents faced in 1987. Thus, the elasticities may change when the demographic and educational structure, the tax system or the constraints change. We refer to Blomquist (1988) for a discussion of the interpretation of labour supply elasticities.

The own-wage elasticities for males are numerically small which is in accordance with the findings in almost all labour supply studies, for instance as found in similar applications to Swedish (Aaberge *et al.*, 1990) and Norwegian (Aaberge *et al.*, 1995) data. Wage elasticities for

Table IV. Aggregate labour supply elasticities^a

Type of elasticity		Male elasticities		Female elasticities	
		Own-wage elasticities	Cross-elasticities	Own-wage elasticities	Cross-elasticities
Elasticity of the probability of participation	Uncompensated	0.046 (0.001)	-0.081 (0.002)	0.654 (0.006)	-0.357 (0.008)
	Compensated	0.043 (0.002)	-0.065 (0.005)	0.640 (0.029)	-0.141 (0.017)
Elasticity of the conditional expectation of total supply of hours	Uncompensated	0.007 (0.001)	-0.035 (0.002)	0.078 (0.003)	-0.136 (0.002)
	Compensated	0.054 (0.004)	-0.033 (0.003)	0.096 (0.008)	-0.058 (0.005)
Elasticity of the unconditional expectation of total supply of hours	Uncompensated	0.053 (0.002)	-0.116 (0.002)	0.737 (0.006)	-0.489 (0.008)
	Compensated	0.098 (0.004)	-0.098 (0.005)	0.742 (0.034)	-0.198 (0.017)

^a Standard deviations in parentheses.

Table V. Conditional aggregate labor supply elasticities^a for male and female members of the 10% poorest, the 10% richest and the 80% 'middle-class' households

Type of elasticity		Male elasticities		Female elasticities	
		Own-wage elasticities	Cross-elasticities	Own-wage elasticities	Cross-elasticities
Elasticity of the probability of participation	Uncompensated	I 0.053 (0.005)	-0.109 (0.006)	2.837 (0.127)	-1.089 (0.097)
		II 0.051 (0.001)	-0.086 (0.002)	0.742 (0.007)	-0.356 (0.009)
		III -0.010 (0.002)	-0.013 (0.001)	0.031 (0.004)	-0.122 (0.004)
	Compensated	I 0.052 (0.011)	-0.084 (0.018)	9.621 (1.100)	2.767 (0.586)
		II 0.047 (0.002)	-0.064 (0.004)	0.646 (0.030)	-0.198 (0.016)
		III 0.004 (0.004)	-0.054 (0.014)	0.082 (0.018)	-0.057 (0.016)
Elasticity of the conditional expectation of total supply of hours	Uncompensated	I 0.021 (0.004)	-0.017 (0.009)	0.467 (0.061)	-1.410 (0.590)
		II 0.011 (0.001)	-0.045 (0.002)	0.100 (0.003)	-0.150 (0.002)
		III -0.030 (0.002)	-0.015 (0.002)	0.004 (0.003)	-0.060 (0.003)
	Compensated	I -0.176 (0.023)	0.002 (0.011)	1.261 (0.250)	-0.226 (0.177)
		II 0.046 (0.003)	-0.039 (0.004)	0.111 (0.009)	-0.150 (0.002)
		III 0.015 (0.006)	-0.014 (0.007)	0.014 (0.007)	-0.008 (0.003)
Elasticity of the unconditional expectation of total supply of hours	Uncompensated	I 0.075 (0.008)	-0.126 (0.008)	3.441 (0.189)	-1.454 (0.111)
		II 0.062 (0.002)	-0.130 (0.002)	0.832 (0.007)	-0.501 (0.008)
		III -0.041 (0.003)	-0.029 (0.003)	0.035 (0.004)	0.181 (0.005)
	Compensated	I 0.229 (0.030)	-0.082 (0.002)	12.086 (1.316)	2.459 (0.560)
		II 0.093 (0.004)	-0.103 (0.005)	0.764 (0.036)	-0.263 (0.015)
		III 0.019 (0.007)	-0.067 (0.018)	0.096 (0.024)	-0.066 (0.017)

Notes:

I = 10% poorest household.

II = 80% in the middle of the distribution of disposable household income.

III = 10% richest household.

^a Standard deviations in parentheses.

females are significantly higher than for males. This is especially the case for the elasticity of the probability of participation with respect to wages.

A remarkable result is the strong cross-effects, which neutralize the own-wage effects. If all wages are raised by 1%, our results indicate that the unconditional expectation of labour supply among men is nearly unaffected (reduced by 0.06%). The females net response is a modest increase of 0.25%. This result should be kept in mind when we later present the results of some policy simulations. This result can also explain why labour supply estimates based on time series give numerically small and often insignificant wage elasticities.

Table V reports aggregate labour supply elasticities conditional on income. The purpose is to examine how the labour supply responses for females and males vary with household income. A striking result in Table V is that wage elasticities are declining with household income. This result, however, is in accordance with what is previously found for Sweden and Norway (cf. Aaberge *et al.*, 1990, 1995). For the poorest households we find very strong wage responses among females, but these responses are sharply decreasing with increasing income. We also note that among the poorest females the income elasticity is numerically quite high. The income elasticities can be derived from the uncompensated and compensated wage elasticities.

Only in one of the specified groups do we find a negative uncompensated elasticity (males belonging to 10% of the richest households). Thus, in this group the negative income effect of wage changes dominate and make the male labour supply curve bend (slightly) backwards.

The comparison of our results with previous results based on Italian data is problematic for the reasons mentioned in the Introduction. Previous estimates are based on local samples or on data sets created by merging different samples. None of those previous studies model the partners' decisions as simultaneous. Moreover, a less precise approximation of the tax system is used in previous studies. Table VI reports the uncompensated elasticities estimated by Colombino and Del Boca (1990) for married women. As one should expect, the results of Table VI suggest that the standard Hausman approach tends to exaggerate the hours response among workers, especially with respect to own wage.

Table VI. Elasticities for married women (from Colombino and Del Boca, 1990)

	Wage	Income
Participation	0.64	-0.44
Conditional hours	0.54	-0.22
Unconditional hours	1.18	-0.66

6. POLICY SIMULATIONS

The empirical estimates of the households' preferences and their opportunity sets allow us to perform policy simulations (changes in tax rates), provided that the opportunity densities are kept fixed. Recall that the opportunity measures of offered hours are assumed to be determined by institutional constraints and/or firm-specific hours of work regulations. These constraints are not likely to change as a consequence of say, changes in the tax system.

One purpose of the simulation experiments is to examine the influence of certain tax reforms on labour supply, income levels and income inequality among households (married couples with or without children).

The basic income concepts are gross income (Y) and disposable income (equal to consumption C) defined as:

$$Y = w_F h_F + w_M h_M + I_1 + I_2 \quad (36)$$

and

$$C = Y - S(w_F h_F, w_M h_M, I_1) \quad (37)$$

where I_1 and I_2 are taxable and non-taxable non-labour family income, respectively, and S is the tax function.

To perform the simulation experiments the following procedure is used: Draw r points,

$$\{h_F(t), h_M(t), \varepsilon(t)\} \quad t = 1, 2, \dots, r$$

where $\{h_F(t)\}$ and $\{h_M(t)\}$ are drawn from uniform distributions with full-time peaks and $\{\varepsilon(t)\}$ are drawn from distribution (3).

Find the realized hours ($h_F(t)$ and $h_M(t)$), given the wages (w_F , w_M) by maximizing

$$\psi(h_F(t), h_M(t), w_F, w_M) + \varepsilon(t)$$

with respect to $t = 1, 2, \dots, r$. Repeat this procedure for every household in the sample. When r is large, this procedure yields results that are close to an 'exact' simulation of the model.

Table VII reports the results of three simulations. The first is a reference case in which the factual tax rules (as of 1987) are used to give the model predictions of participation rates, expected annual hours of work (given participation), gross earnings, gross family income, taxes and disposable income.

Table VII. Participation rates, annual hours of work, gross income, taxes and disposable income (1000 LIT) for couples under three different tax regimes. Means

Tax	Participation		Expected annual hours of work, given participation		Gross earnings		Gross income	Taxes	Disposable income
	F	M	F	M	F	M		Households	
1987 tax rules	0.40	0.96	690	1932	7,300	23,600	36,600	9,400	27,300
Proportional taxes ^a	0.38	0.96	667	1952	7,200	24,300	37,200	9,400	27,800
Increased progression ^a	0.36	0.96	625	1933	6,700	23,700	36,100	9,400	26,800

^a The proportional tax rate 26.3% and the progressive tax rates on gross earnings are derived under the restriction of constant tax revenue equal to the revenue under the 1987 rules.

In the second simulation the 1987 taxes are replaced by a proportional tax on wage earnings. The third simulation experiment aims at examining the effect of a more progressive tax structure. The tax rates in the eight tax brackets of the existing system are increased proportionally. In both cases total tax revenue is kept fixed. The proportional tax case ends up with a tax rate of 26.3%. In the third case, more progressive tax rates are applied in order to generate a degree of redistribution of 0.86. The degree of redistribution is measured by the ratio between the Gini coefficients of the distributions of disposable and gross income.

Table VIII. The Gini coefficient^a for distributions of households gross and disposable income, and degree of redistribution under various tax regimes

Tax regime	Gross income	Disposable income	Degree of redistribution
1987 tax rules	0.253 (0.001)	0.238 (0.001)	0.94
Proportional taxes	0.247 (0.001)	0.247 (0.001)	1.00
Increased progression	0.255 (0.001)	0.220 (0.001)	0.86

^a Standard deviations in parentheses.

Table VII indicates that the effects of these tax changes are remarkably weak. A shift to proportional taxes has a negative impact on female labour supply, and a positive, but weak, impact on male labour supply. Strong cross-effects and weak own-wage effects among middle- and high-income earners are the main reason for the stability in aggregate labour supply. The reform characterized by increased progression has a weak negative net effect on both female and male labour supply.

Table VIII provides estimates of the Gini coefficient in the distributions of gross and disposable family income. A shift to a proportional tax makes the distribution of pre-tax income slightly more equal, but this effect is not strong enough to counteract the removal of the redistributive effect provided by the 1987 tax rules. As a result we observe that the inequality in the distribution of disposable income increases by 4%. The case of increased progression reveals no effect on inequality in the pre-tax income distribution, but reduces after-tax income inequality by 7.6%.

7. CONCLUSIONS

The present study employs a particular econometric framework which allows for complex non-convex budget sets, highly non-linear labour supply curves and imperfect markets with institutional constraints. The empirical results show that male labour supply, on average, is rather inelastic while labour supply among females, especially participation, is an order of magnitude more elastic. Other important findings are strong cross-effects and wage elasticities that are sharply declining with family income.

Policy simulations indicate that tax changes which affect the disposable income of all family members will have rather modest impacts on aggregate labour supply and on the distribution of disposable income among married couples. These results are due to strong cross-effects and rather inelastic labour supply among individuals with middle and high incomes.

Although the present analysis may be considered as an extension of previous labour supply studies there are several aspects that can be improved on in future work. First, it would be important to collect data on more job attributes than wage rates and hours of work (see Tinbergen, 1956; Hartog, 1987). Second, the assumption of non-correlated offered hours and wage rates should be subject to an empirical test. However, this requires independent observations of wage rates and hours of work, preferably at the firm level. Third, since the observed peaks in the hours of work distribution are due partly to firm technology and partly to the results of central negotiations, it would be of interest to collect and apply firm-specific and institutional information

on how hours are set. Moreover, it would also be interesting to know whether changing distributional assumptions will yield different policy conclusions.

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